

CAPE BYRON LH, Australia

WMO: 945990

Lat: 28.633S

Lon: 153.633E

Elev: 322

StdP: 14.53

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	50.6	52.1	29.0	23.3	57.2	32.4	27.2	57.8	33.0	62.8	29.6	62.1	17.1	240	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	85.2	73.9	83.2	73.3	81.6	72.6	76.9	81.1	75.6	80.0	74.6	79.1	12.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.7	136.0	78.8	74.4	130.1	78.0	73.3	125.3	77.3	40.6	80.9	39.4	80.1	38.4	79.3	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.3	28.4	25.1	47.4	89.4	1.5	1.8	46.4	90.7	45.5	91.7	44.7	92.7	43.6	94.0		
(5)			40.0	78.9	1.6	2.0	38.8	80.4	37.9	81.5	36.9	82.7	35.8	84.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	75.9	75.4	73.8	69.6	64.7	61.1	60.0	61.7	65.2	67.9	70.8	73.4		
	DBStd	6.41	2.99	3.11	3.14	3.26	3.31	3.39	3.54	3.68	3.60	3.47	3.24	3.32		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	507	0	0	0	4	47	123	158	115	41	15	3	1		
	CDD50	6662	802	712	739	587	456	333	311	362	456	555	625	724		
	CDD65	1694	337	292	274	141	37	7	4	12	47	105	178	260		
	CDH74	6121	1778	1516	1045	232	20	0	2	8	39	139	409	933		
	CDH80	749	282	205	108	4	0	0	0	0	3	5	23	119		
(14) Wind	WSAvg	13.8	14.1	15.1	14.1	13.9	12.6	13.7	12.7	13.2	13.9	14.4	14.0	14.3		
(15) Precipitation	PrecAvg	68.1	6.3	6.9	8.5	7.6	7.8	6.4	3.8	3.4	2.3	4.4	5.1	6.0		
	PrecMax	97.5	17.5	16.1	25.7	24.1	21.5	16.2	13.0	10.9	11.1	22.3	13.7	16.8		
	PrecMin	44.3	1.2	1.6	1.2	1.8	0.7	0.6	0.0	0.4	0.0	0.3	0.7	1.7		
	PrecStd	15.4	3.9	3.2	5.6	4.5	4.6	4.3	3.0	2.7	2.3	3.6	3.1	3.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	87.4	85.4	80.5	76.2	72.5	72.0	75.0	78.2	80.4	82.6	86.2		
		MCWB	75.9	74.9	74.9	71.7	67.1	64.0	62.2	60.6	63.2	67.8	71.0	72.7		
	2%	DB	85.4	84.4	82.8	78.3	74.0	70.1	69.8	71.5	74.4	77.4	80.1	83.3		
		MCWB	74.3	74.1	73.4	70.2	65.7	62.9	60.7	61.7	63.2	67.0	69.3	72.1		
	5%	DB	82.9	82.4	80.9	76.7	72.1	68.4	68.0	69.6	72.4	75.3	78.0	80.8		
		MCWB	73.4	73.9	72.3	69.2	64.5	61.9	60.5	60.9	63.3	66.5	69.0	71.5		
	10%	DB	81.1	80.8	79.0	75.0	70.5	66.8	66.3	67.9	70.8	73.5	76.2	78.8		
		MCWB	72.9	73.1	71.5	68.3	63.4	60.6	59.5	60.0	63.0	66.1	68.5	71.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.6	79.5	77.2	75.3	71.5	67.8	66.6	66.8	69.4	72.3	74.5	76.8		
		MCDB	83.4	82.0	81.8	77.5	73.3	69.4	69.0	69.6	71.7	74.9	78.9	80.9		
	2%	WB	76.7	77.0	75.4	72.8	68.0	65.3	64.8	64.8	67.5	70.5	72.5	75.0		
		MCDB	81.2	80.7	79.9	76.1	72.2	68.2	67.4	68.8	70.5	73.6	76.4	79.7		
	5%	WB	75.5	75.6	74.2	71.0	66.2	63.6	62.8	63.3	66.3	69.0	71.1	73.7		
		MCDB	79.9	80.0	78.5	74.6	70.2	66.9	66.1	67.8	69.9	72.7	75.0	77.9		
	10%	WB	74.5	74.4	73.0	69.5	64.6	62.0	60.9	61.7	64.8	67.9	70.1	72.5		
		MCDB	78.9	79.2	77.3	73.7	69.0	65.4	64.8	66.6	69.1	71.9	74.1	76.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	9.6	9.4	9.5	9.8	9.3	10.2	10.8	10.5	10.4	10.4	10.0		
		MCDBR	13.2	12.2	11.9	11.0	10.9	10.3	11.7	12.3	12.5	13.4	13.1	13.6		
		MCWBR	6.3	6.1	6.1	6.5	7.5	7.5	8.5	8.8	8.3	8.1	6.7	7.4		
		MCDBR	11.2	10.7	10.6	10.0	9.6	9.0	9.7	10.7	9.8	11.3	10.9	11.1		
	5% WB	MCWBR	6.2	5.8	6.5	6.9	7.1	7.5	8.2	8.7	7.0	7.3	6.3	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.384	0.370	0.352	0.338	0.316	0.310	0.301	0.314	0.349	0.378	0.375	0.385		
	taud		2.523	2.575	2.622	2.636	2.654	2.662	2.675	2.620	2.514	2.458	2.510	2.505		
	Ebn at Noon		304	303	298	288	281	275	283	292	294	297	305	304		
	Edh at Noon		36	33	30	27	24	22	23	27	33	36	36	36		
(44) All-Sky Solar Radiation	RadAvg		2063	1873	1594	1336	1101	910	1058	1342	1670	1873	2042	2047		
	RadStd		160	147	86	68	65	59	85	104	134	161	158	207		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (21 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	-104	+308	+189	(47)

Nomenclature: See separate page